

**CUSTOMER SEGMENTATION
MARKETING STRATEGY
WITH USE OF
STATISTICAL AND UNSUPERVISED LEARNING
METHODS**

Agata Griffin

1st October 2025

Contents

PROBLEM STATEMENT	3
METHODS	3
CUSTOMER SEGMENTATION BASED ON DATA ANALYSIS	3
CLUSTER ANALYSIS AND RECOMMENDATIONS.....	7
Limitations and Considerations	9

PROBLEM STATEMENT

A fine-tuned marketing strategy is a baseline of business profitability, customer acquisition and brand loyalty. This analysis performs customer base segmentation based on their purchasing behaviour.

The purpose of this analysis is to:

- Define the optimal number of clusters to aid marketing and sales strategy
- Describe customer characteristics given their cluster definition
- Provide assessment and critique of segmentation techniques used.

METHODS

The following methods are applied for customer analysis and customer segmentation.

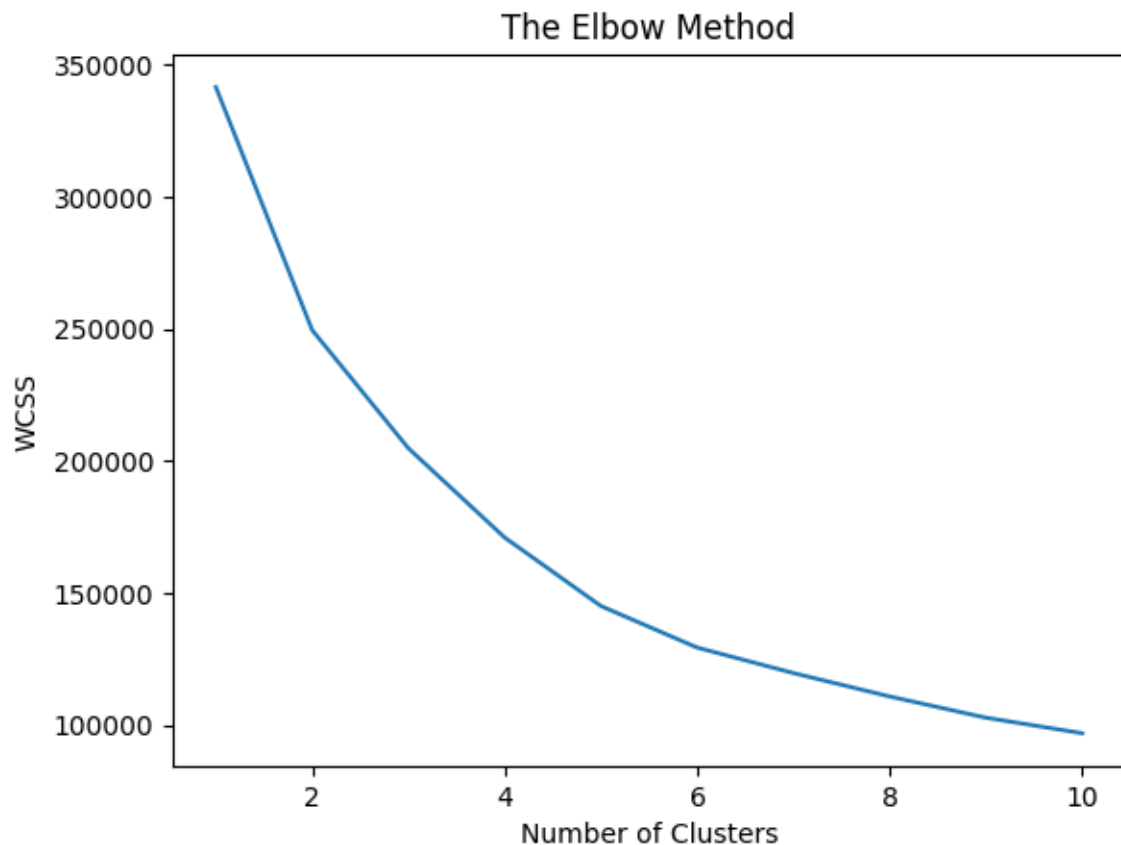
Method	Model Output Parameters	Method Outline
Machine Learning (ML)		
• Elbow Method	Graphical representation of optimal k-value(s)	Identify a point of minimum rate of inertia decrease equivalent highest individual clusters compactness
• Silhouette Method	Vector of Silhouette coefficients	Measures inter-cluster cohesion and intra-cluster separateness
• Hierarchical Dendrogram	Number of pronounced dendrogram branches	Deterministic method with implementation dependent on computational resources

CUSTOMER SEGMENTATION BASED ON DATA ANALYSIS

Customer segmentation was performed with k-means. In order to apply k-means technique, three separate methodologies were used to determine the optimal number of clusters as an input to k-means: Elbow, Silhouette Score and Hierarchical Dendrogram.

Elbow Method

As can be seen on the below chart, the rate of decrease of WCCS slows down at $n=2$, and further at $n=5$ and $n=6$, indicating possible levels of k-means.



Silhouette Score

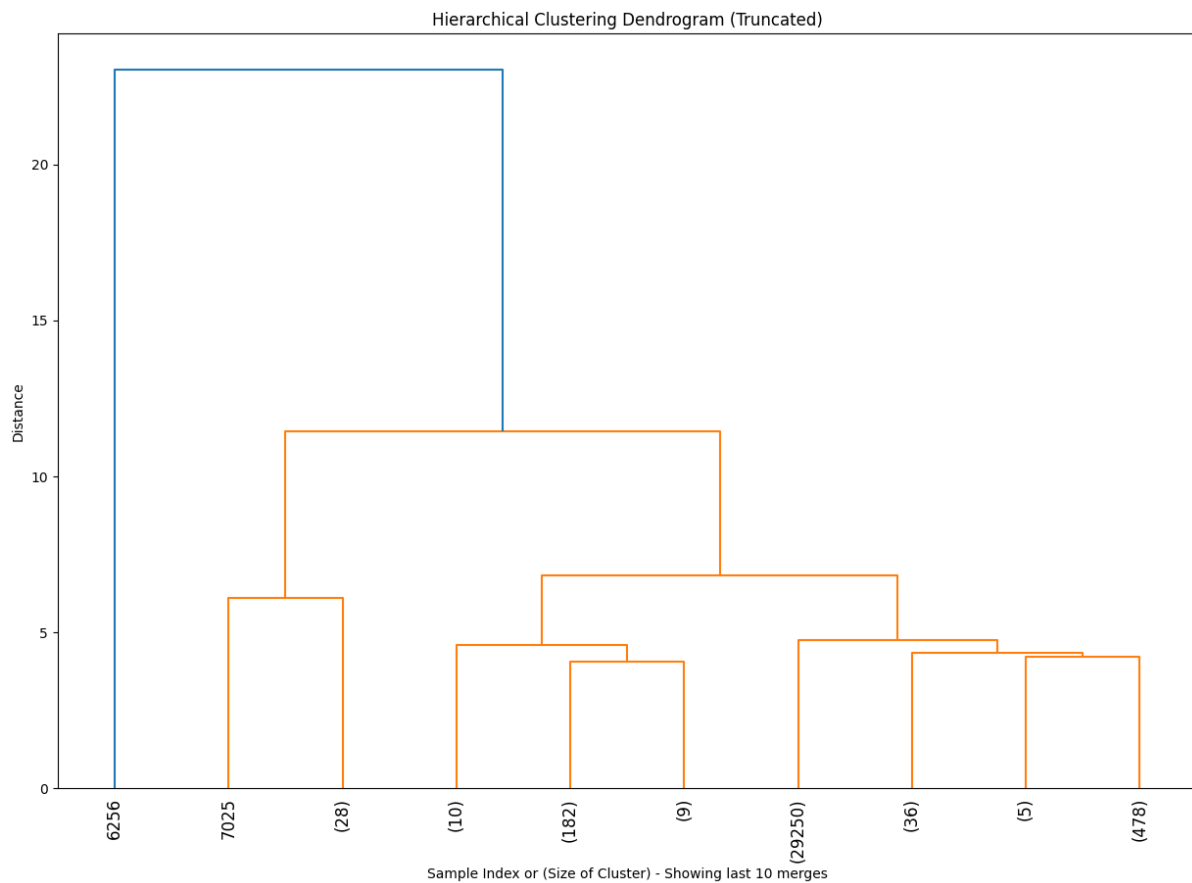
The model output of potential k-means indicated that the optimal number of customer clusters is 5.

n-clusters	2	3	4	5	6	8	10
avg silhouette_score	0.2629	0.2526	0.2532	0.2668	0.2526	0.2344	0.2253

Silhouette score is a more effective method of deriving k-means compared to Elbow method, as in addition to minimising intra-cluster distances, it looks to maximise the intra-cluster distances, with a coefficient vector output reducing ambiguity of results interpretation, and allowing attributing cluster label to each data point.

Hierarchical dendrogram

Although, a hierarchical dendrogram was generated based on a sample of 30,000 data points (given computational limitations), certain observations can be made. The longer branches of the dendrogram indicate cluster distances, and their total equates to the optimal number of clusters. Based on the generated dendrogram, the potential optimal number of clusters can be defined at k=2 or k=4.

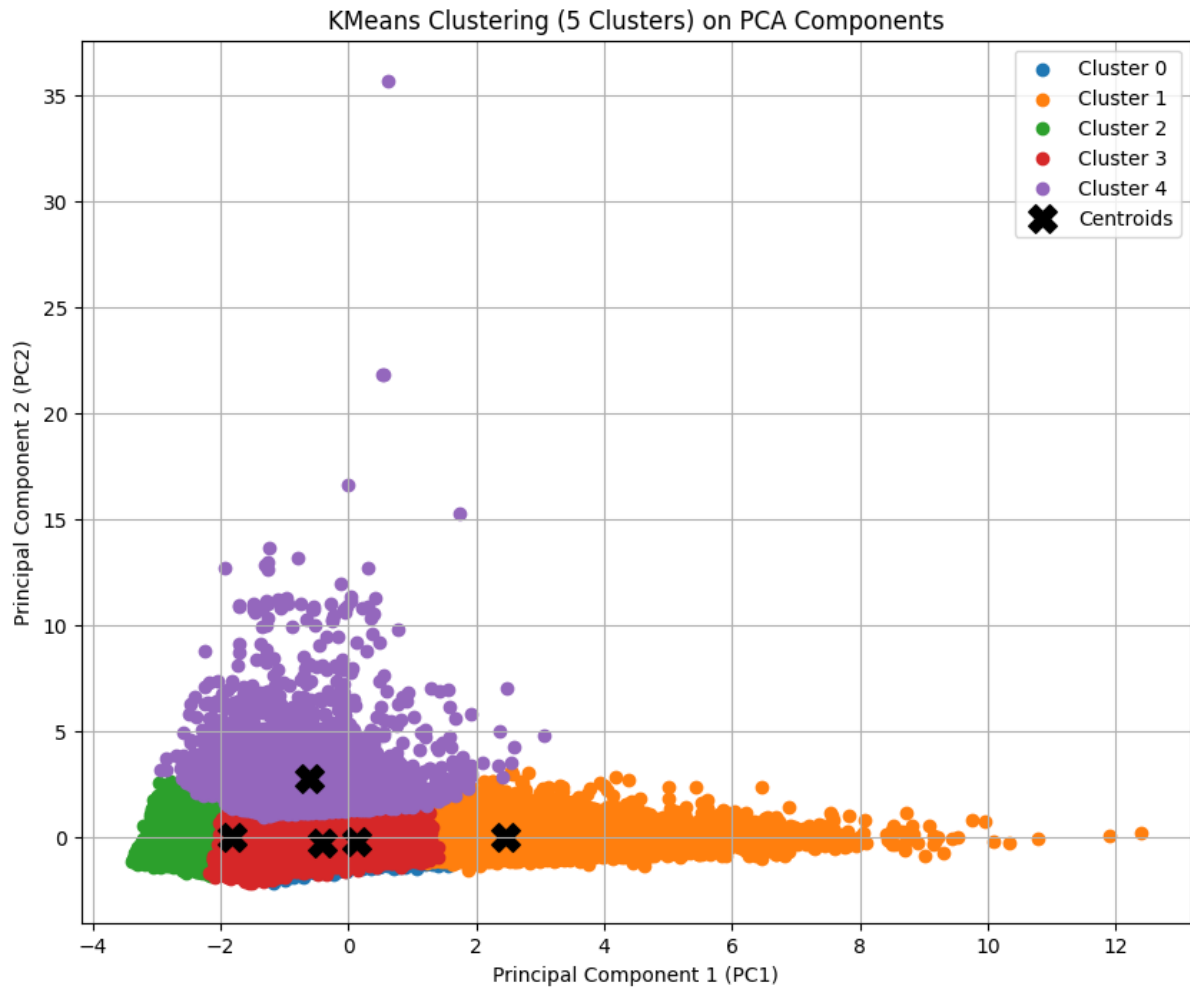


In conclusion, given the limited effectiveness of the Hierarchical Dendrogram method due to computational constraints, as both Silhouette method and Elbow method indicated $n=5$ as the optimal number of clusters, it is proposed to set it as k-means model input.

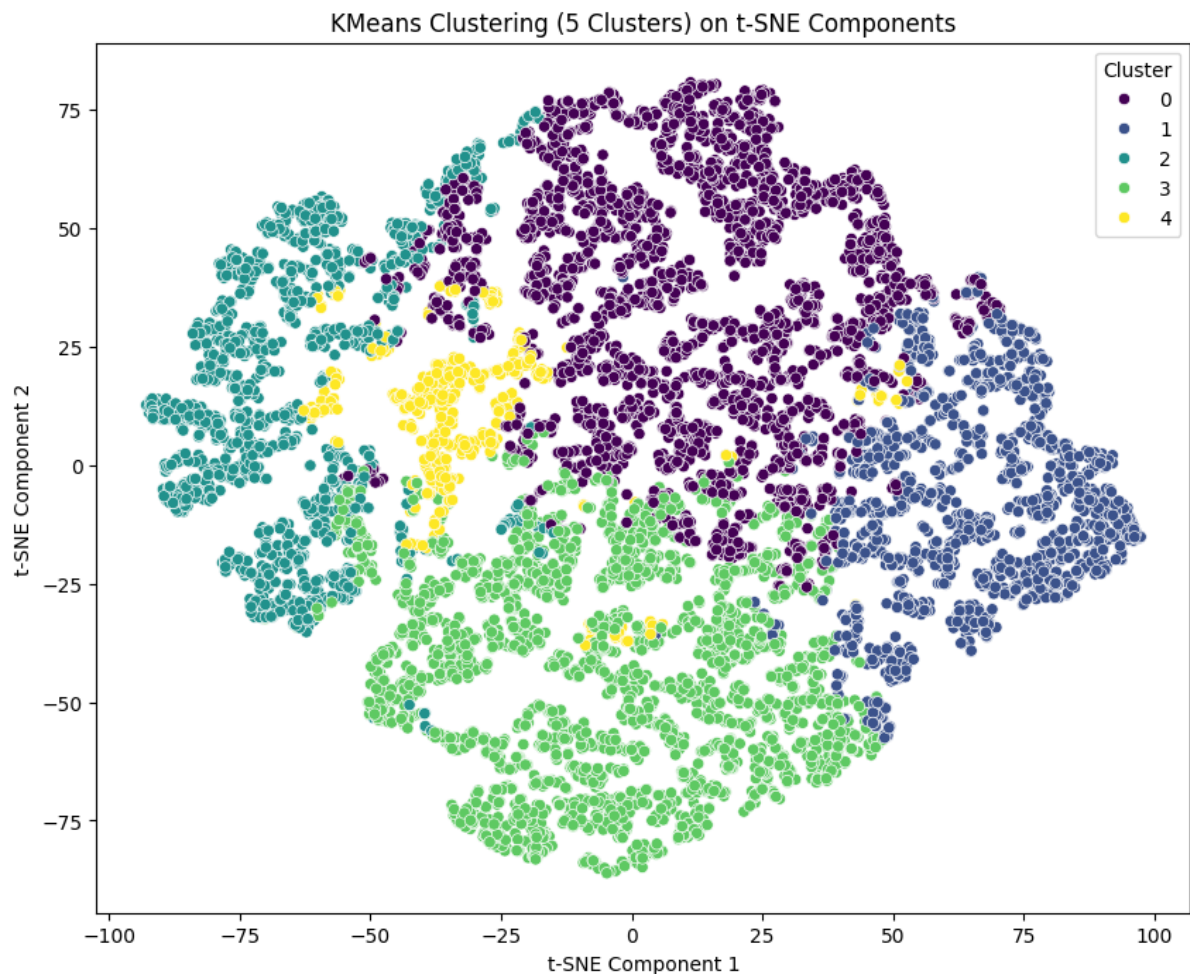
k-means analysis

k-means generates a vector of cluster labels categorising each of the data points. Cluster_0, cluster_1, cluster_2, cluster_3 and cluster 4 are generated corresponding to $k = 5$.

Initial analysis with PCA, shows 5 clusters and their centroids.



The PCA cluster analysis reveals that the clusters, whilst compact, are not separated. t-SNE analysis is performed to validate customer segmentation based on 5 clusters. This analysis is based on a sample of 10,000 customers, given computational limitations.



t-SNE analysis improves the presentation of customer segments compared to PCA. Whilst the clusters are distinct, they are clearly separated in space.

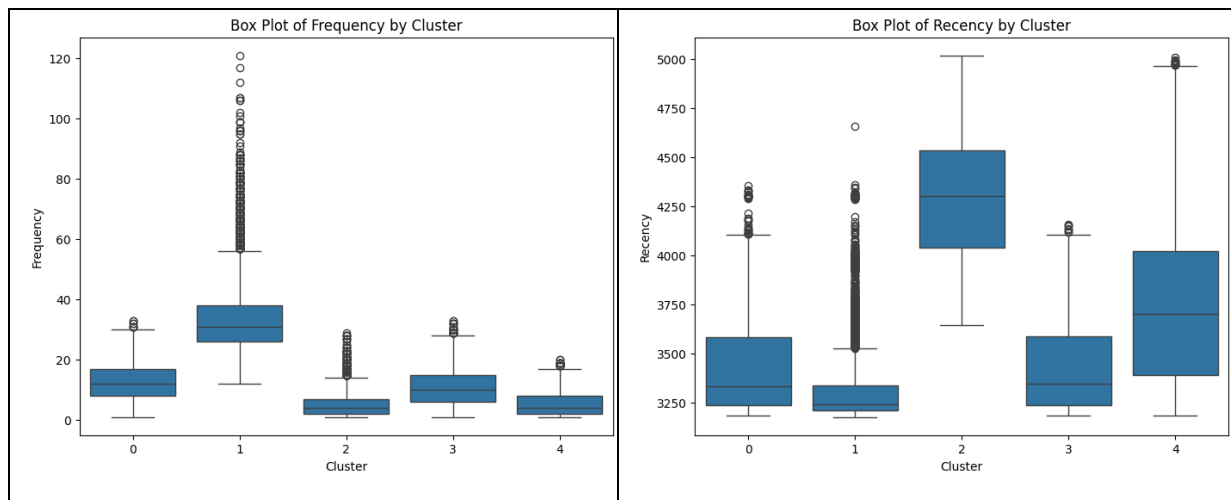
CLUSTER ANALYSIS AND RECOMMENDATIONS

Based on customer cluster analysis by 5 features: Frequency, Recency, CLV, Avg Unit Cost and Age, it is observed that:

“New buyers”: Cluster_0 represents an active customer group with an average recency and relatively high frequency of purchases, with some positive outliers. CLV is relatively high given moderate recency, indicating customers who like the product, purchase it frequently and at an average price. If this group continues their purchasing pattern, this will contribute positively to the company’s bottom line. **Recommendation:** Marketing effort, if any, should be focused on building brand loyalty.

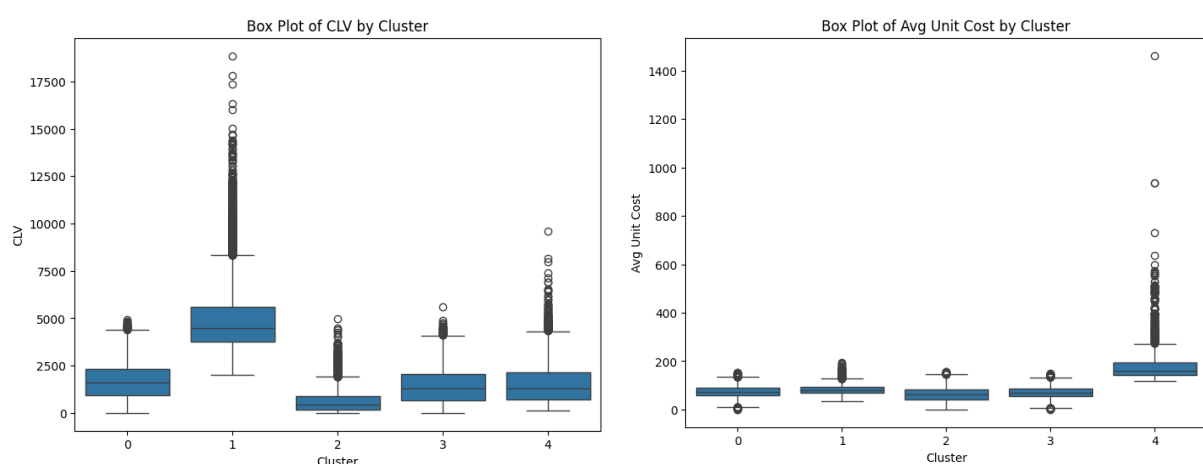
“Loyal customers”. Cluster 1 is characterised by the highest frequency of purchases with the lowest recency lowest (albeit with positive outliers) and highest CLV (with very many positive outliers), the average unit cost is second-highest median with a tight IQR. This indicates customers

who are not price sensitive, purchase the product frequently and recently who have the highest CLV. **Recommendation:** ensure customer retention.



“Dissatisfied customers”. Customers in Cluster 2 have highest recency, lowest frequency of purchases, and low CLV. It represents customers who might have tried the product, possibly in response to a promotion, but have not repeated the purchase, perhaps as they are not satisfied with it or cannot afford to purchase it regularly. **Recommendation:** market research to understand why customers did not repeat the product purchase.

“Discerning Customers”. Customers in Cluster 3 are potentially sensitive to prices as they have negative avg unit cost outliers (as well as positive), relatively low recency with medium frequency and an average CLV. As this cluster has the highest median customer age, this might indicate discerning customers who know the product they want and who might not respond to a marketing campaign but may be sensitive to price discounts. **Recommendation:** marketing strategy based on product offers.



“High Net-worth Customers”. Cluster 4 represents customers who are not price sensitive with highest avg unit cost combined with a very large number of positive outliers. The frequency of purchases is low combined with the second highest median recency and the widest IQR. **Recommendation:** marketing effort aimed at an increased customer engagement

Limitations and Considerations

Ethical consideration should be given when clustering customers based on their age group, as this can be a protected characteristic.